

AT1 Data Interpretation Report

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Q1: Employee Salary Analysis

Problem:

Employee Salary Analysis

Code:

```
import pandas as pd
```

```
df = pd.read_csv("employees.csv")
```

```
print("Employees earning more than 50000")
```

```
print(df[df["Salary"] > 50000])
```

```
print("\nAverage Salary Department-wise")
```

```
print(df.groupby("Department")["Salary"].mean())
```

Output:

```
===== RESTART  
Employees earning more than 50000  
   Emp_ID  Name Department  Salary  
1     102   Bob         IT     60000  
2     103 Charlie    Finance    75000  
3     104  David         IT     52000  
5     106  Frank    Finance    90000  
  
Average Salary Department-wise  
Department  
Finance      82500.0  
HR           42500.0  
IT           56000.0  
Name: Salary, dtype: float64
```

Conclusion:

The employee dataset was successfully read from the CSV file. Employees earning more

than ₹50,000 were displayed, and the average salary for each department was calculated using Pandas.

Q4: Sorting Dataset

Problem:

Sorting Dataset

Code:

```
import pandas as pd
```

```
df = pd.read_csv("sales.csv")
```

```
sorted_df = df.sort_values(by="Sales", ascending=False)
```

```
print("Top 10 Highest Selling Products")
```

```
print(sorted_df.head(10))
```

Output:

```
=====
Top 10 Highest Selling Products
  Product  Sales
10      TV  98000
6      Phone  95000
0     Laptop  85000
8     Camera  70000
3    Monitor  65000
5     Tablet  55000
9     Watch  40000
4     Printer  30000
7     Speaker  20000
2   Keyboard  18000
```

Conclusion:

The sales dataset was successfully sorted in descending order based on sales amount. The top 10 highest-selling products were displayed accurately.

Q6: Data Cleaning

Problem:

Data Cleaning

Code:

```
import pandas as pd
```

```
df = pd.read_csv("employee_clean.csv")
```

```
df["Name"] = df["Name"].str.strip().str.title()
```

```
df["Phone"] = df["Phone"].fillna("Not Available")
```

```
print(df)
```

Output:

```
=====
```

	Name	Phone
0	Alice	9876543210.0
1	Bob	Not Available
2	Charlie	9123456789.0
3	David	Not Available

Conclusion:

The employee dataset was successfully cleaned by removing extra spaces, converting names to title case, and replacing missing phone numbers with "Not Available", resulting in a consistent and clean dataset.

Q8: Data Aggregation

Problem:

Data Aggregation

Code:

```
import pandas as pd
```

```
df = pd.read_csv("monthly_sales.csv")
```

```
print("Total Sales =", df["Sales"].sum())
```

```
print("Average Sales =", df["Sales"].mean())
```

```
print("Maximum Sales =", df["Sales"].max())
```

```
print("Minimum Sales =", df["Sales"].min())
```

Output:

```
===== RESTART: C:/Users/Harshini Darshan/OneDrive/astl.
Total Sales = 200000
Average Sales = 33333.333333333336
Maximum Sales = 50000
Minimum Sales = 20000
```

Conclusion:

The monthly sales data was successfully analyzed to calculate the total, average, maximum, and minimum sales using Pandas aggregation functions

Q10: Merge Two CSV Files

Problem:

Merge Two CSV Files

Code:

```
import pandas as pd
```

```
emp = pd.read_csv("employees1.csv")
```

```
dept = pd.read_csv("departments.csv")
```

```
merged = pd.merge(emp, dept, on="Dept_ID")
```

```
print(merged)
```

Output:

```
===== RESTART: C:/Users/Harshini Darshan/OneDrive/astl.
  Emp_ID  Name  Dept_ID Department
0    101  Alice        1         HR
1    102   Bob        2         IT
2    103 Charlie        1         HR
3    104  David        3    Finance
```

Conclusion:

The employee and department datasets were successfully merged using the Department ID, producing a combined dataset with complete employee and department information.

Q12: Convert CSV into JSON

Problem:

Convert CSV into JSON

Code:

```
import pandas as pd
```

```
df = pd.read_csv("student.csv")
```

```
df.to_json("student.json", orient="records", indent=4)
```

```
print("CSV converted to JSON successfully")
```

Output:

student		15-07-2026 14:53	Microsoft Excel Co...	1 KB
student		15-07-2026 14:53	JSON Source File	1 KB

```
CSV converted to JSON successfully
```

Conclusion:

The CSV file was successfully converted into JSON format using Pandas, demonstrating efficient data format transformation.

Q13: JSON Data Filtering

Problem:

JSON Data Filtering

Code:

```
import pandas as pd
```

```
df = pd.read_json("employee.json")
```

```
print("Salary > 60000")
```

```
print(df[df["Salary"] > 60000])
```

```
print("\nIT Department")
```

```
print(df[df["Department"] == "IT"])
```

Output:

```
===== RESTART: C:/U:
Salary > 60000
   Name Department  Salary
1    Bob          IT   65000
2  Charlie          IT   72000

IT Department
   Name Department  Salary
1    Bob          IT   65000
2  Charlie          IT   72000
```

Conclusion:

The JSON dataset was successfully read and filtered to display employees with salaries greater than ₹60,000 and employees belonging to the IT department.

Q17: Data Interpretation using Python

Problem:

Data Interpretation using Python

Code:

```
marks = [65,78,90,45,88,72,35]
```

```
print("Highest Mark =", max(marks))
```

```
print("Lowest Mark =", min(marks))
```

```
print("Average Mark =", sum(marks)/len(marks))
```

```
passed = 0
```

```
for i in marks:
```

```
    if i >= 40:
```

```
        passed += 1
```

```
print("Students Passed =", passed)
```

Output:

```
Highest Mark = 90
Lowest Mark = 35
Average Mark = 67.57142857142857
Students Passed = 6
```

Conclusion:

Student marks were successfully analyzed using Python. The highest mark, lowest mark, average mark, and the number of students who passed were calculated correctly.

Q18: Dictionary-Based Analysis

Problem:

Dictionary-Based Analysis

Code:

```
employees = {
    "Alice":45000,
    "Bob":70000,
    "Charlie":55000,
    "David":80000
}

highest = max(employees, key=employees.get)
lowest = min(employees, key=employees.get)

average = sum(employees.values())/len(employees)

print("Highest Salary Employee =", highest)
print("Lowest Salary Employee =", lowest)
print("Average Salary =", average)
```

Output:

```
===== RESTART: C:/Users/Harshini I  
Highest Salary Employee = David  
Lowest Salary Employee = Alice  
Average Salary = 62500.0
```

Conclusion:

Employee salary information stored in a dictionary was successfully analyzed to identify the highest-paid employee, the lowest-paid employee, and the average salary.

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